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Statistical estimation and nonlinear filtering in environmental pollution

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Abstract
Motivated by the water pollution detection, this paper studies a nonlinear filtering problem over an infinite time interval. The signal to be estimated, which indicates the concentration of undesired chemical in a river, is driven by a stochastic partial differential equation involves unknown parameters. Based on discrete observation, strongly consistent estimators of unknown parameters are derived at first. With the optimal filter given by Bayes formula, the uniqueness of invariant measure for the signal-filter pair has been verified. The paper then establishes approximation to the optimal filter with estimators, showing that the pathwise average distance, per unit time, of the computed approximating filter from the optimal filter converges to zero in probability. Simulation results are presented at last.

Keywords Nonlinear filtering · Stochastic partial differential equation · Optimal filter · Invariant probability measure · Pathwise average distance

1 Introduction

This work is concerned with approximations to a nonlinear filtering problem over an infinite time interval. The model is a river pollution model studied by Chapter 7 of Kallianpur and Xiong (1995): In a domain $[0, 1]$, some undesired chemical is releasing from an unknown location β behaving as a Poisson process with unknown intensity λ. Let $U(t, x)$ denote the

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time evolution of the concentration of undesired chemical at time t and location x. Then $U(t,x)$ is governed by the following stochastic partial differential equation (SPDE):

$$\partial_t U(t,x) = a\partial_x^2 U(t,x) + b\partial_x U(t,x) - \alpha U(t,x) + \partial_t N_{\lambda t}\delta_\beta(x) \tag{1}$$

given the initial $U(0,\cdot) \geq 0$ and Neumann boundary conditions:

$$\partial_x U(t,0) = 0, \quad \partial_x U(t,1) = 0, \qquad t \geq 0,\ x \in [0,1].$$

The setting here is considered in dimension $d = 1$. The dispersion coefficient $a \in \mathbb{R}_+$, drift velocity of river flow $b \in \mathbb{R}$, dilution rate of chemical $\alpha \in \mathbb{R}_+$ are all given. N_t is a standard Poisson process independent of $U(0,\cdot)$ and δ_β is the Dirac measure at β. Suppose the unknown parameter $\theta \equiv (\beta,\lambda)$ takes value in a bounded set $\Theta = \{(\beta,\lambda); 0 < \beta < 1, 0 < \lambda \leq M\}$. We assume the other parameters are known just for simplicity of notation and our main motivation is to find the location of the polluter and the intensity of the pollution. In fact, we can also deal with the case that other parameter are unknown. We refer to Remark 2 for more details.

Note that SPDE (1) is understood as follows. Let $0 = \tau_1 < \tau_2 < \cdots$ be the arrival times of the Poisson point process $N_{\lambda t}$. For each fixed ω, we solve SPDE (1) for $t \in [\tau_j, \tau_{j+1})$, $j = 0,1,2,\cdots$, by the following partial differential equation (PDE):

$$\begin{cases} \partial_t U(t,x) = a\partial_x^2 U(t,x) + b\partial_x U(t,x) - \alpha U(t,x) \\ \partial_x U(t,0) = 0,\ \partial_x U(t,1) = 0, \\ U(\tau_i,x) = U(\tau_i-,x) + \delta_\beta(x). \end{cases}$$

It is well-known from the theory of PDE that the above PDE has a classical solution $U(t,x)$ for $\tau_j < t < \tau_{j+1}$. Note that

$$\mathbb{P}\left(\exists\, j,\ \text{s.t. } t = \tau_j\right) = 0.$$

Thus, $\{U(t,x) : t \geq 0,\ x \in [0,1]\}$ can be regarded as a random field, although $U(\tau_j,\cdot)$, as a measure on $[0,1]$, does not have a density. The unique solution of SPDE (1) will be shown as formula (5) in the next section. Its proof as well as more details of the model are discussed in Chapter 7 of Kallianpur and Xiong (1995).

Define $\mathcal{X}$ as a Polish space of finite measures on the Borel σ-field of $[0,1]$. For convenience, we regard U as an $\mathcal{X}$-valued process given by

$$U_t(A) = \int_A U(t,x)dx, \qquad \forall A \in \mathcal{B}[0,1].$$

For simplicity of notation, $U_{t_j}(j = 1,2,\cdots)$ will be denoted by U_j in the following.

The concentration of undesired chemical can rarely be observe directly. Instead, an observation station is set up and the chemical concentration around the station is measured subject to random error. Herein, we consider the following observation model:

$$Y_i = h(\langle U_i, \phi_0\rangle) + \varepsilon_i, \qquad i = 1,2,\cdots \tag{2}$$

where $\langle U_i, \phi_0\rangle = \int_0^1 \phi_0(x)U(t_i,x)dx$, h is a bounded continuous function on $\mathbb{R}$. ϕ_0 can be either bounded continuous function or Dirac delta function on $[0,1]$. ε_i's are independent standard normal distributed random variables independent of N_t, $t_i = i\Delta$ with $\Delta > 0$.

The observation filtration $\mathcal{F}_j^Y = \sigma\{Y_i; i = 1,\cdots,j\}$ contains all the available information about β, λ, U_t. The primary aim of our filtering problem is to get an estimate of β, λ based on $\mathcal{F}_j^Y$ in L^2-space, and approximate the optimal filter

$$\langle \Pi_j, \phi\rangle = \mathbb{E}\{\phi(U_j) \mid \mathcal{F}_j^Y\}, \quad \forall \phi \in C_b(\mathcal{X}). \tag{3}$$

It is easy to show that the conditional expectation $\mathbb{E}\{\phi(U_j) \mid \mathcal{F}_j^Y\}$ is the best estimate of $\phi(U_j)$ based on $\mathcal{F}_j^Y$ under square loss.

Nonlinear filtering is widely used in engineering, natural science, machine learning and even mathematical finance; see for examples, Van Leeuwen (2010), Bishop (2006), Brigo and Hanzon (1998). For the approximations to nonlinear filtering problems over the past few decades, some approaches have been developed. The pioneering work is Markov chain approximation raised by Kushner (1977). The idea is to find a Markov chain which suitably approximated the actual signal process, then to construct the approximating filter for the Markov chain and actual observations. Another algorithm to solve the nonlinear filtering problems is the particle system introduced by Gordon et al. (1993). The solution is represented by a system of weighted particles whose positions and weights are governed by SDEs that can be solved numerically; see Chapter 8 of Xiong (2008) for details. There are also some other numerical schemes coming out meanwhile, such as projection filter and assumed density approach; see Brigo et al. (1999) and Kushner (2011). To approximate the signal process and optimal filter for model (1)-(2), the current paper will construct Markov processes with the idea of Markov chain approximation and the obtained estimators of β, λ. Note that the problem is of interest over a very long time interval. Thus when talking about the error of approximated filter from optimal filter, we would use the pathwise average per unit time rather than the mean value; following the remarkable works done by Budhiraja and Kushner (1999), Budhiraja and Kushner (2001). Kouritzin et al. (2004) also studied a nonlinear filtering problem over a long time interval. Their signal process was a reflecting diffusion in a random environment. In this article, the signal to be estimated is driven by an SPDE involved with jump process and unknown parameters, leading to a different way to construct its approximating filter process.

It should be highlighted that this article will prove in a systematic way that the signal-filter pair has a unique probability invariant measure. The critical importance of uniqueness of the joint process was first raised by Kushner and Huang (1986). In this paper, Theorems 1 and 4 will present a rigorous proof that the signal process has a unique invariant measure and the filter forgets its initial condition, yielding uniqueness of invariant measure of optimal filter by Theorem 7.1 in Budhiraja and Kushner (1999).

The article is organized as follows. In the next section, we are to look for the estimators of λ and β based on the data set $\{Y_i; i = 1, \cdots, n\}$. Section 3 will construct an approximating filter with the obtained estimators and discuss the pathwise average errors over an infinite time interval. Simulation for the estimators as well as approximating filter are given in Sect. 4. The article is concluded in Sect. 5 by some remarks.

2 Statistical estimation

This section focuses on estimating parameters β, λ. There are some studies on parameter estimation for SPDEs based on discrete observations in time and space. Huebner and Rozovskii (1995) found out the maximum likelihood estimators in parabolic SPDEs with Dirichlet conditions. Hildebrandt and Trabs (2021) discovered moment estimators under Dirichlet boundary conditions in an SPDE similar to formula (1) whereas the jump process is replaced by cylindrical Brownian motion. Both works did not consider measurement noise.

The current model investigates a few parameters in an SPDE involved with Poisson jump process under Neumann boundary conditions. Signal process is measured subject to random error. This section is going to take the method of moment estimate. It is easy to implement and can provide consistent estimators which are of use to establish approximated filter for process $U(t, x)$.

Let $\mathcal{P}(\mathcal{X})$ be the collection of all Borel probability measures on Polish space $\mathcal{X}$. Denote the law of U_t by $\mathcal{L}(U_t)$. Here is a theorem about the convergence of $\mathcal{L}(U_t)$, with which the limits of sample moments will be derived.

Theorem 1 *As $t \to \infty$, the sequence of measures $\mathcal{L}(U_t)$ converges in $\mathcal{P}(\mathcal{X})$ to a probability measure F which is the unique invariant probability measure of the process U_t. Especially, for any bounded continuous function f, we have*

$$\frac{1}{n}\sum_{i=1}^{n} f(Y_i) \xrightarrow{a.s.} \int_{\mathcal{X}}\int_{\mathbb{R}} f(h(\langle \nu, \phi_0\rangle) + x)\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}} F(d\nu)dx, \quad as\ n \to \infty. \tag{4}$$

Furthermore, F is characterized by its characteristic functional

$$\mathbb{E}\exp\left(i\int_{\mathcal{X}} \langle \nu, f\rangle\, F(d\nu)\right) = \exp\left(\lambda\int_0^{\infty}\left(e^{ie^{-\alpha s}T_s f(\beta)} - 1\right)ds\right),$$

for any $f \in C_b([0, 1])$, where $\{T_t\}$ is the semigroup generated by $L = a\partial_x^2 + b\partial_x$ with Neumann boundary conditions.

Proof From Theorem 7.2.1 of Kallianpur and Xiong (1995), Eq. (1) has a unique solution given by

$$U(t, x) = e^{-\alpha t}\int_0^1 P_t(y, x)U_0(dy) + \int_0^t e^{-\alpha(t-s)}P_{t-s}(\beta, x)dN_{\lambda s} \tag{5}$$

where $P_t(y, x)$ is the Green's function of the operator $L = a\partial_x^2 + b\partial_x$ with Neumann boundary conditions given by

$$P_t(y, x) = \sum_{j=0}^{\infty} e^{-\sigma_j t}\psi_j(y)\psi_j(x)$$

and

$$\begin{aligned}
&\sigma_0 = 0, \quad \sigma_j = a\left(c^2 + (j\pi)^2\right);\\
&\psi_0(x) = \sqrt{\frac{2c}{1 - e^{-2c}}}, \quad \psi_j(x) = \sqrt{2}e^{cx}\sin(j\pi x + k_j);\\
&k_j = \tan^{-1}\left(-\frac{j\pi}{c}\right), \quad c = -\frac{b}{2a}, \quad j = 1, 2, \cdots
\end{aligned}$$

Now, we prove the tightness of $\{\mathcal{L}(U_t)\}$ in $\mathcal{P}(\mathcal{X})$. To this end, it is enough to prove that for any bounded continuous function f on $[0, 1]$, the family $\{\mathcal{L}(\langle U_t, f\rangle)\}$ is tight in $\mathcal{P}(\mathbb{R})$. Without loss of generality, take $0 \le f \le K$, and write $T_t f(y) = \int_0^1 P_t(y, x)f(x)dx$. Then

$$\begin{aligned}
\mathbb{E}\langle U_t, f\rangle &= e^{-\alpha t}\int_0^1 T_t f(y)U_0(dy) + \mathbb{E}\int_0^t e^{-\alpha(t-s)}T_{t-s}f(\beta)dN_{\lambda s}\\
&\le Ke^{-\alpha t}\int_0^1 U_0(dy) + \int_0^t e^{-\alpha(t-s)}K\lambda ds\\
&\le K\int_0^1 U_0(dy) + K,
\end{aligned}$$

which indicates the tightness of $\{\mathcal{L}(\langle U_t, f\rangle)\}$. By Prokhorov theorem, tightness implies relative compactness, meaning that each subsequence of $\{\mathcal{L}(U_t)\}$ contains a further subsequence converging weakly. In this way, if the characteristic function of $\{\langle U_t, f\rangle\}$ converges to a unique function, then one can conclude that all subsequences of $\{\mathcal{L}(\langle U_t, f\rangle)\}$ converges to the same distribution. In fact,

$$\begin{aligned}
\mathbb{E}\exp\left(i\left\langle U_t, f\right\rangle\right) &= \mathbb{E}\exp\left(ie^{-\alpha t}\int_0^1 T_t f(y)U_0(dy) + i\int_0^t e^{-\alpha(t-s)}T_{t-s}f(\beta)dN_{\lambda s}\right)\\
&= \exp\left(ie^{-\alpha t}\int_0^1 T_t f(y)U_0(dy)\right)\\
&\quad\times\exp\left(\lambda\int_0^t\left(\exp\left(ie^{-\alpha(t-s)}T_{t-s}f(\beta)\right)-1\right)ds\right)\\
&= \exp\left(ie^{-\alpha t}\int_0^1 T_t f(y)U_0(dy)\right)\exp\left(\lambda\int_0^t\left(e^{ie^{-\alpha s}T_s f(\beta)}-1\right)ds\right)\\
&\to \exp\left(\lambda\int_0^\infty\left(e^{ie^{-\alpha s}T_s f(\beta)}-1\right)ds\right), \text{ as } t\to\infty.
\end{aligned}$$

This implies the uniqueness of the limited measure, that is, $\mathcal{L}(U_t)\to F$ as $t\to\infty$, and hence the transformation of the Markov chain U_t is uniquely ergodic. In consequence, formula (4) follows by the Birkhoff Ergodic Theorem. □

Denote $\theta_0 \equiv (\beta_0, \lambda_0)$ as the true values of θ. Let

$$g_1(\theta) = \int_{\mathcal{X}} h(\langle \nu, \phi_0\rangle)F(d\nu), \qquad g_2(\theta) = \int_{\mathcal{X}} h^2(\langle \nu, \phi_0\rangle)F(d\nu),$$

and the Jacobian matrix

$$J(\theta) = \begin{bmatrix} \partial g_1(\theta)/\partial\beta & \partial g_1(\theta)/\partial\lambda \\ \partial g_2(\theta)/\partial\beta & \partial g_2(\theta)/\partial\lambda \end{bmatrix}.$$

Theorem 2 *Let $m_r(n) = n^{-1}\sum_{i=1}^n Y_i^r$, $r = 1, 2$. Suppose $g_r(\theta)$ posses continuous partial derivative on Θ and the Jacobian* $\det(J(\theta))$ *be different from zero everywhere on Θ. If the following equations*

$$\begin{aligned} m_1(n) &= g_1(\theta)\\ m_2(n) &= g_2(\theta) + 1\end{aligned}$$

have a unique solution, denoted by $\hat{\theta}_n = (\hat{\beta}_n, \hat{\lambda}_n)$, with probability 1 *as $n\to\infty$, then this solution is a strongly consistent estimator for* (β_0, λ_0).

Proof Let S be the image of the set Θ under the continuous mapping $g = (g_1, g_2 + 1)$. With probability 1 the inverse mapping g^{-1} is locally one-to-one and is a continuous mapping from S into Θ. Denote $m(n) = (m_1(n), m_2(n))$. From Theorem 1, we have

$$m(n) \overset{a.s.}{\to} g(\theta_0), \quad \text{as } n\to\infty.$$

Hence for large n with probability 1, the point $m(n)\in S$; in this case, equations $m(n) = g(\theta)$ are solvable and their solution is necessarily of the form $\hat{\theta}_n = g^{-1}(m(n))\in\Theta$. As a result, thanks to the continuity of g^{-1}, we have

$$\hat{\theta}_n = g^{-1}(m(n)) \overset{a.s.}{\to} \theta_0, \quad \text{as } n\to\infty,$$

and finish the proof. □

Remark 1 In addition to Dominated Convergence Theorem and bounded condition for θ, it yields that $\hat{\theta}_n$ convergences in L^1-norm to θ_0.

Remark 2 The above method adapts to the cases with more unknown parameters. For example if the drift α and β, λ are to be estimated, in addition to g_1, g_2, define $g_3(\theta) = \int_{\mathcal{X}} h^3(\langle \nu, \phi_0 \rangle) F(d\nu), \theta \equiv (\alpha, \beta, \lambda)$ and $m_3(n) = n^{-1} \sum_{i=1}^n Y_i^3$. With one more equation $m_3(n) = g_3(\theta) + 3g_1(\theta)$ in Theorem 2, a strongly consistent estimator for θ_0 is derived.

With the obtained estimators, now we are able to construct the Markov processes to efficiently approximate the optimal filter $\Pi.$ in (3).

3 Stochastic filtering

Let $\mathcal{F}_j^N = \sigma\{N_u; 0 \le u \le t_j, t_j = j\Delta\}$ and $\mathcal{F}_j = \mathcal{F}_j^N \vee \mathcal{F}_j^Y$. We take simplified notations $Y_{0,j} = \{Y_i; i = 1, \cdots, j\}$ and $U_{0,j} = \{U_{t_i}; i = 1, \cdots, j\}$. For each $j \in \mathbb{N}$, let

$$R(U_{0,j}, Y_{0,j}) = \exp\left\{\sum_{i=1}^{j} [Y_i h(\langle U_i, \phi_0 \rangle) - \frac{1}{2} h^2(\langle U_i, \phi_0 \rangle)]\right\}.$$

Then, one can define a probability measure $\mathbb{Q}$ by

$$\left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{\mathcal{F}_j} = R^{-1}(U_{0,j}, Y_{0,j}).$$

Let $\mathbb{E}_{\mathbb{Q}}$ denote the expectation with respect to $\mathbb{Q}$. Unless further notice, $\mathbb{E}$ denote the expectation with respect to the original probability measure $\mathbb{P}$. In the following theorem, we have an expression for the optimal filter $\Pi.$ in terms of $\mathbb{Q}$.

Theorem 3 *i) Under $\mathbb{Q}$, $\{Y_1, \cdots, Y_j\}$ are mutually independent standard normal random variables and are independent of $(N_s)_{0 \le s \le t_j}$.*
ii) Under $\mathbb{Q}$, $(N_s)_{0 \le s \le t_n}$ has the same distribution as under $\mathbb{P}$.
iii) (The Bayes formula) For any bounded and continuous function ϕ,

$$\begin{aligned}\mathbb{E}\{\phi(U_j)|\mathcal{F}_j^Y\} &= \frac{\mathbb{E}_{\mathbb{Q}}\{\phi(U_j)\frac{d\mathbb{P}}{d\mathbb{Q}}|\mathcal{F}_j^Y\}}{\mathbb{E}_{\mathbb{Q}}\{\frac{d\mathbb{P}}{d\mathbb{Q}}|\mathcal{F}_j^Y\}} \\ &= \frac{\mathbb{E}_{\mathbb{Q}}\{\phi(U_j)R(U_{0,j}, Y_{0,j})|\mathcal{F}_j^Y\}}{\mathbb{E}_{\mathbb{Q}}\{R(U_{0,j}, Y_{0,j})|\mathcal{F}_j^Y\}}. \end{aligned} \tag{6}$$

The proof is done by the similar arguments used in Theorems 3.1 and 3.2 of Mandal and Mandrekar (2000).

Let $\tilde{U}_0$ be a measure-valued random variable with the same distribution as U_0 but to be independent of U_0 and all other processes. Let $\tilde{N}$ be a standard Poisson process independent of all other processes. Define

$$\tilde{U}(t, x) = e^{-\alpha t} \int_0^1 P_t(y, x)\tilde{U}_0(dy) + \int_0^t e^{-\alpha(t-s)} P_{t-s}(\beta_0, x) d\tilde{N}_{\lambda_0 s}. \tag{7}$$

Thanks to Theorem 3, the conditional expectation (6) can be rewritten as

$$\langle \Pi_j, \phi \rangle = \frac{\mathbb{E}_{\{\Pi_0, Y_{0,j}\}}\{\phi(\tilde{U}_j)R(\tilde{U}_{0,j}, Y_{0,j})\}}{\mathbb{E}_{\{\Pi_0, Y_{0,j}\}}\{R(\tilde{U}_{0,j}, Y_{0,j})\}},$$

where $\mathbb{E}_{\{\Pi,Y\}}G(\tilde{U}_t, Y)$ denotes the conditional expectation of a function G of $\tilde{U}_t$ and Y, given the data Y and initial distribution of $\tilde{U}_t$ is Π_0. In the end, owing to the Markov property of U_t, the optimal filter satisfies the semigroup relation

$$\langle \Pi_j, \phi\rangle = \frac{\mathbb{E}_{\{\Pi_{j-1},Y_j\}}\left\{\phi(\tilde{U}_1)R(\tilde{U}_1, Y_j)\right\}}{\mathbb{E}_{\{\Pi_{j-1},Y_j\}}\left\{R(\tilde{U}_1, Y_j)\right\}}, \tag{8}$$

on which we impose a definition

$$R(\tilde{U}_1, Y_j) = \exp\left\{Y_j h(\langle \tilde{U}_1, \phi_0\rangle) - \frac{1}{2}h^2(\langle \tilde{U}_1, \phi_0\rangle)\right\}.$$

In (8), Π_{j-1} is the distribution of $\tilde{U}_0$.

The following is a critical important theorem of this paper.

Theorem 4 *The signal-filter pair $\{U_j, \Pi_j\}$ has a unique invariant probability measure.*

Proof Note that U_j is a hidden Markov process. We first verify that

$$\mathbb{E}\left(\|\mathbb{P}(U_j \in \cdot|U_0) - \mu_0\|_{TV}\right) \to 0, \qquad \text{as } j \to \infty, \tag{9}$$

with invariant measure $\mu_0 \equiv \mathbb{P}(V_\infty \in \cdot)$, where $\|\cdot\|_{TV}$ is the total variation norm and

$$V_j = \int_0^{t_j} e^{-\alpha(t_j-s)} P_{t_j-s}(\beta, \cdot) dN_{\lambda s}.$$

Note that for each j, $\{N_{\lambda s}, 0 \le s \le t\}$ and $\{N_{\lambda t_j} - N_{\lambda(t_j-s)}\}$ are equal in distribution and hence,

$$\begin{aligned} V_j &\overset{\mathcal{D}}{=} \int_0^{t_j} e^{-\alpha(t_j-s)} P_{t_j-s}(\beta, \cdot) d\left(N_{\lambda t_j} - N_{\lambda(t_j-s)}\right) \\ &= \int_0^{t_j} e^{-\alpha s} P_s(\beta, \cdot) dN_{\lambda s} \\ &\to V_\infty \equiv \int_0^{\infty} e^{-\alpha s} P_s(\beta, \cdot) dN_{\lambda s}. \end{aligned}$$

Note that $U_j - V_j \to 0$. Let (U_j) be extended to $j \in \mathbb{Z}$ as a stationary process and let $\mathbb{P}^x$ be a version of regular conditional probability $\mathbb{P}^{U_0} = \mathbb{P}(\cdot|U_0)$. Since the limit of U_j does not depend on the initial, it is easy to verify that for any $A \in \cap_{j\le 0}\mathcal{F}^U_{-\infty,j}$ and $x, y \in \mathcal{X}$, $\mathbb{P}^x(A) = \mathbb{P}^y(A) \in \{0, 1\}$. It then follows from the remark after Conjecture 1.7 of Van Handel (2012) or Theorem 2.3 of Van Handel (2009) that (9) holds.

Further, the non-degenerate property (cf. Definition 1.5 of Van Handel (2012)) of the observations follows from (2) directly. Hence, by Theorem 1.6 of Van Handel (2012) that the exchangeability (1.1) of that paper holds.

One can see that the limit of U_t, as t goes to ∞, is independent of the initial value U_0, so the filter Π_j forgets its initial condition. Thus, the signal-filter pair $\{U_j, \Pi_j\}$ has a unique probability invariant measure by Theorem 7.1 of Budhiraja and Kushner (1999). □

Here we proceed to the establishments of approximating signal and the computed approximating filter. Set

$$\tilde{U}^n(t, x) = e^{-\alpha t}\int_0^1 P_t(y, x)\tilde{U}_0(dy) + \int_0^t e^{-\alpha(t-s)} P_{t-s}(\hat{\beta}_n, x) d\tilde{N}_{\hat{\lambda}_n s}, \tag{10}$$

and $\tilde{U}^n_t \equiv \tilde{U}^n(t,\cdot)$. Define the approximating filter Π^n recursively by

$$\langle \Pi^n_j, \phi\rangle = \frac{\mathbb{E}_{\{\Pi^n_{j-1},Y_j\}}\big\{\phi(\tilde{U}^n_1)R(\tilde{U}^n_1,Y_j)\big\}}{\mathbb{E}_{\{\Pi^n_{j-1},Y_j\}}\big\{R(\tilde{U}^n_1,Y_j)\big\}}. \tag{11}$$

The initial filter Π^n_0 coincides with the distribution of U_0 which is given. The next theorem shows the convergence property of approximating signal $\tilde{U}^n_1$.

Theorem 5 *For any bounded continuous function ϕ,*

$$\mathbb{E}\big|\langle\tilde{U}^n_1,\phi\rangle - \langle\tilde{U}_1,\phi\rangle\big|^2 \to 0, \qquad as\ n\to\infty.$$

Proof Note that for any bounded continuous function ϕ,

$$\begin{aligned}
&\mathbb{E}\big|\langle\tilde{U}^n_1,\phi\rangle - \langle\tilde{U}_1,\phi\rangle\big|^2\\
&\quad= \mathbb{E}\Big|\int_0^{t_1} e^{-\alpha(t_1-s)}\int_0^1 \phi(x)P_{t_1-s}(\hat\beta_n,x)dxd\tilde{N}_{\hat\lambda_n s}\\
&\qquad- \int_0^{t_1} e^{-\alpha(t_1-s)}\int_0^1 \phi(x)P_{t_1-s}(\beta_0,x)dxd\tilde{N}_{\lambda_0 s}\Big|^2\\
&\quad\le 2\mathbb{E}\Big|\int_0^{t_1} e^{-\alpha(t_1-s)}\int_0^1 \phi(x)\Big[P_{t_1-s}(\hat\beta_n,x)dx - P_{t_1-s}(\beta_0,x)\Big]dxd\tilde{N}_{\hat\lambda_n s}\Big|^2\\
&\qquad+ 2\mathbb{E}\Big|\int_0^{t_1} e^{-\alpha(t_1-s)}\int_0^1 \phi(x)P_{t_1-s}(\beta_0,x)dxd\Big[\tilde{N}_{\hat\lambda_n s} - \tilde{N}_{\lambda_0 s}\Big]\Big|^2\\
&\quad\le K\mathbb{E}\hat\lambda_n(1+t_1\hat\lambda_n)\int_0^{t_1} e^{-2\alpha(t_1-s)}\int_0^1\Big|P_{t_1-s}(\hat\beta_n,x)dx - P_{t_1-s}(\beta_0,x)\Big|^2dxds\\
&\qquad+ 2\int_0^{t_1} e^{-2\alpha(t_1-s)}\Big(\int_0^1 \phi(x)P_{t_1-s}(\beta_0,x)dx\Big)^2\mathbb{E}\big|\hat\lambda_n-\lambda_0\big|\big(1+t_1\big|\hat\lambda_n-\lambda_0\big|\big)ds.
\end{aligned}$$

Obviously, the second term on the RHS goes to zero with Remark 1. For the double integral,

$$\begin{aligned}
&\int_0^{t_1}\int_0^1\Big|P_{t_1-s}(\hat\beta_n,x)dx - P_{t_1-s}(\beta_0,x)\Big|^2dxds\\
&\quad\le \int_0^1 2\sum_{j=0}^\infty\int_0^{t_1} e^{-2\sigma_j(t_1-s)}ds\psi_j^2(x)\Big|e^{c\hat\beta_n}\sin(j\pi\hat\beta_n+k_j) - e^{c\beta_0}\sin(j\pi\beta_0+k_j)\Big|^2dx\\
&\quad\le 8e^{2c}\sum_{j=0}^\infty\frac{1}{2\sigma_j}(1-e^{-2\sigma_j t_1})\Big(\big|e^{c\hat\beta_n}-e^{c\beta_0}\big|^2\\
&\qquad+ 2e^{2c\beta_0}\Big|\sin\frac{j\pi(\hat\beta_n-\beta_0)}{2}\cos\frac{j\pi(\hat\beta_n+\beta_0)+2k_j}{2}\Big|^2\Big)\\
&\quad\le K\sum_{j=0}^\infty\frac{1}{\sigma_j}\Big(\big|e^{c\hat\beta_n}-e^{c\beta_0}\big|^2 + \Big|\sin\frac{j\pi(\hat\beta_n-\beta_0)}{2}\Big|^2\Big)\\
&\quad\le \sum_{j=0}^\infty\frac{K}{a(c^2+(j\pi)^2)}\Big(\big(e^{c\delta^*}(\hat\beta_n-\beta_0)\big)^2 + \big(j\pi|\hat\beta_n-\beta_0|\big)^{\frac12}\Big)\\
&\quad\le \sum_{j=0}^\infty\frac{K}{a(c^2+(j\pi)^2)}\Big((\hat\beta_n-\beta_0)^2 + \big(j\pi|\hat\beta_n-\beta_0|\big)^{\frac12}\Big),
\end{aligned}$$

where $\delta^* = \delta^*(\beta_0, \hat{\beta}_n)$ falls between β_0 and $\hat{\beta}_n$. With Remark 1, we obtain that the double integral converges to 0 in L^1-norm. $\square$

We now need to investigate the asymptotic behavior of $\{U_j, \Pi_j^n\}$ and prove the main result. Recall that in Theorem 4, $\{U_j, \Pi_j\}$ is shown to has a unique invariant measure, denoted by ρ. Let f be any bounded continuous real-valued function. In what follows, we will show that

$$\frac{1}{k}\sum_{j=1}^{k} f(U_j, \Pi_j^n) \to \int_{\mathcal{X}\times\mathcal{P}(\mathcal{X})} f(\nu, \alpha)\rho(d\nu, d\alpha) \tag{12}$$

in probability as n, k go to ∞.

Let $\bar{\mathcal{X}}$ be the compactification of $\mathcal{X}$ which is metrizable. In fact, it can be identified by $[0, 1] \times \mathcal{P}([0, 1])$ since $\mathcal{X}$ can be identified with $[0, \infty) \times \mathcal{P}([0, 1])$, while the one-point compactification of $[0, \infty)$ is identified as $[0, 1]$. Then, any sequence of measures on $\bar{\mathcal{X}}$ is tight.

For each j and n, define

$$B_j = \sum_{i=1}^{j} \varepsilon_i, \quad \Psi^n(j, \cdot) = (U_{j+\cdot}, \Pi^n_{j+\cdot}, Y_{j+\cdot}, B_{j+\cdot} - B_j),$$

where $\{G_{j+\cdot}\}$ means the process G_t at time $j+1, j+2, \cdots$. Let $I_C(\Psi^n(j, \cdot))$ denote the indicator function of the event $\Psi^n(j, \cdot) \in C$ and C be any measurable set in the product path space of $\Psi^n(j, \cdot)$. Here we define the occupation measures:

$$Q^{n,k}(C) = \frac{1}{k}\sum_{j=1}^{k} I_C\big(\Psi^n(j, \cdot)\big) \tag{13}$$

that take value in the space of probability measures on the product path space

$$\mathcal{P}\big(\mathcal{S}(\bar{\mathcal{X}}; \mathbb{N}) \times \mathcal{S}(\mathcal{P}(\bar{\mathcal{X}}); \mathbb{N}) \times \mathcal{S}(\mathbb{R}; \mathbb{N}) \times \mathcal{S}(\mathbb{R}; \mathbb{N})\big).$$

$\mathcal{S}(E; \mathbb{N})$ is the path space of E-valued functions. To verify formula (12), in the sequel, we will discuss the tightness and weak convergence of the measure-valued random variables $Q^{n,k}(\cdot)$.

Theorem 6 *$\{U_{j+\cdot}; j \geq 0\}$ is tight in $\mathcal{S}(\bar{\mathcal{X}}, \mathbb{N})$; $\{\Pi^n_{j+\cdot}; n > 0, j \geq 0\}$ is tight in $\mathcal{S}(\mathcal{P}(\bar{\mathcal{X}}), \mathbb{N})$; $\{Y_{j+\cdot}; j \geq 0\}$ is tight in $\mathcal{S}(\mathbb{R}, \mathbb{N})$; and $\{Q^{n,k}; n > 0, k > 0\}$ is tight on $\mathcal{P}\big(\mathcal{S}(\bar{\mathcal{X}}; \mathbb{N}) \times \mathcal{S}(\mathcal{P}(\bar{\mathcal{X}}); \mathbb{N}) \times \mathcal{S}(\mathbb{R}; \mathbb{N}) \times \mathcal{S}(\mathbb{R}; \mathbb{N})\big)$.*

Proof (i) $\{U_{j+\cdot}; j \geq 0\}$ is tight since $\bar{\mathcal{X}}$ is compact.

(ii) Obviously, $\{\Pi^n_{j+\cdot}; n > 0, j \geq 0\}$ is tight in $\mathcal{S}(\mathcal{P}(\bar{\mathcal{X}}), \mathbb{N})$ since this space is compact.

(iii) $\{Y_{j+\cdot}; j \geq 0\}$ is tight from the boundedness of h and stationarity of ε_j.

(iv) The sequence $\{\mathbb{E}[Q^{n,k}(\cdot)]\}$ is tight from the tightness of $\{U_{j+\cdot}; j \geq 0\}$, $\{\Pi^n_{j+\cdot}; n > 0, j \geq 0\}$, $\{Y_{j+\cdot}; j \geq 0\}$ and $\{B_{j+\cdot} - B_j; j \geq 0\}$. Thus, the sequence $\{Q^{n,k}(\cdot)\}$ is tight by Theorem 1.6.1 of Kushner (1990). $\square$

Thanks to the tightness property, we understand that each subsequence of $\{Q^{n,k}(\cdot)\}$ converges weakly. Before we identify the limits of subsequences, define the occupation measures for the signal-filter pair

$$Q_1^{n,k}(C) = \frac{1}{k}\sum_{j=1}^{k} I_C\big(\Psi_1^n(j, \cdot)\big), \quad \Psi_1^n(j, \cdot) = (U_{j+\cdot}, \Pi^n_{j+\cdot}). \tag{14}$$

Theorem 7 *Let $Q(\cdot)$ denote any weak limit of $\{Q^{n,k}(\cdot)\}$. Let ω be the canonical variable on the probability space on which $Q(\cdot)$ is defined, and denote the samples by Q^{ω}. For each ω, Q^{ω} induces a process*

$$\Psi_{\cdot}^{\omega} = (U_{\cdot}^{\omega}, \Pi_{\cdot}^{\omega}, Y_{\cdot}^{\omega}, B_{\cdot}^{\omega}).$$

For almost all ω, the following hold.

(i) $\Pi_{\cdot}^{\omega}$ *is $\mathcal{P}(\mathcal{X})$-valued.*
(ii) $(U_{\cdot}^{\omega}, \Pi_{\cdot}^{\omega})$ *is stationary.*
(iii) $\{B_{\cdot}^{\omega}\}$ *is the sum of mutually independent $\mathcal{N}(0,1)$ random variables $\{\varepsilon_i^{\omega}\}$ which are independent of $U_{\cdot}^{\omega}$.*
(iv) $Y_{\cdot}^{\omega} = h(\langle U_{\cdot}^{\omega}, \phi_0\rangle) + \varepsilon_{\cdot}^{\omega}$.
(v) $U_{\cdot}^{\omega}$ *has the transition function of $U_{\cdot}$.*
(vi) *for each integer j and $\phi \in C_b(\mathcal{X})$,*

$$\langle \Pi_j^{\omega}, \phi\rangle = \frac{\mathbb{E}_{\{\Pi_0^{\omega}, Y_{0,j}^{\omega}\}}\{\phi(\tilde{U}_j) R(\tilde{U}_{0,j}, Y_{0,j}^{\omega})\}}{\mathbb{E}_{\{\Pi_0^{\omega}, Y_{0,j}^{\omega}\}}\{R(\tilde{U}_{0,j}, Y_{0,j}^{\omega})\}}. \tag{15}$$

Proof Assume the lower case $\psi_{\cdot} = (u_{\cdot}, \pi_{\cdot}, y_{\cdot}, b_{\cdot})$ is the canonical sample path in the space where $\Psi_{\cdot}^{\omega}$ (of all ω) are defined. Some of the ideas to be used in the proof come from Budhiraja and Kushner (1999).

(i) Let $K_\delta = \{\mu \in \bar{\mathcal{X}}; \mu([0,1]) \le 1/\delta\}$. By the definition of the Π_j^n, we have that

$$\Pi_j^n(K_{\delta j}^c) = \frac{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{I_{K_{\delta j}^c}(\tilde{U}_1^n) R(\tilde{U}_1^n, Y_j)\}}{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{R(\tilde{U}_1^n, Y_j)\}}.$$

Let $\tilde{V}_1^n = \int_0^{t_1} e^{-\alpha(t_1-s)} P_{t_1-s}(\hat{\beta}_n, \cdot) d\tilde{N}_{\hat{\lambda}_n s}$. Thus, we have that, for each n and j,

$$\begin{aligned}
&\Pi_j^n(K_{\delta j}^c)\\
&= \int \mathbb{E}\Big[I_{K_{\delta j}^c}\Big(\int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n\Big)\\
&\quad \times R\Big(\int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n, Y_j\Big)\Big| Y_j\Big] \Pi_{j-1}^n(d\nu)\\
&\quad \Big/ \int \mathbb{E}\Big[R\Big(\int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n, Y_j\Big)\Big| Y_j\Big] \Pi_{j-1}^n(d\nu)\\
&\le \int \mathbb{E}\Big[I_{K_{\delta j}^c}\Big(\int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n\Big) \exp\Big\{h\Big(\Big\langle \int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy)\\
&\quad + \tilde{V}_1^n, \phi_0\Big\rangle\Big) Y_j - \frac{1}{2} h^2\Big(\Big\langle \int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n, \phi_0\Big\rangle\Big)\Big\}\Big| Y_j\Big] \Pi_{j-1}^n(d\nu)\\
&\quad \Big/ \Big\{\int \mathbb{E}\Big[\exp\Big\{h\Big(\Big\langle \int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n, \phi_0\Big\rangle\Big) Y_j\\
&\quad - \frac{1}{2} h^2\Big(\Big\langle \int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n, \phi_0\Big\rangle\Big)\Big\}\Big| Y_j\Big] \Pi_{j-1}^n(d\nu)\Big\}\\
&\le c_1 e^{c_2|\varepsilon_j|} \int \mathbb{E}\Big[I_{K_{\delta j}^c}\Big(\int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n\Big)\Big| Y_j\Big] \Pi_{j-1}^n(d\nu)\\
&= c_1 e^{c_2|\varepsilon_j|} \int_{K_{\delta j-1} \cup K_{\delta j-1}^c} \mathbb{E}\Big[I_{K_{\delta j}^c}\Big(\int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}_1^n\Big)\Big| Y_j\Big] \Pi_{j-1}^n(d\nu)
\end{aligned}$$

$$
\begin{aligned}
&= c_1 e^{c_2|\varepsilon_j|} \int_{K_{\delta j-1}\cup K^c_{\delta j-1}} \mathbb{P}\Big\{\Big(\int e^{-\alpha t_1} P_{t_1}(y,\cdot)\nu(dy) + \tilde{V}^n_1\Big) \geq \frac{1}{\delta j}\Big|Y_j\Big\}\Pi^n_{j-1}(d\nu) \\
&\leq c_1 e^{c_2|\varepsilon_j|}\Big\{\delta^j \int_{K_{\delta j-1}} \mathbb{E}\Big[e^{-\alpha t_1}\nu([0,1]) + \tilde{V}^n_1\Big|Y_j\Big]\Pi^n_{j-1}(d\nu) + \Pi^n_{j-1}(K^c_{\delta j-1})\Big\} \\
&\leq c_1 e^{c_2|\varepsilon_j|}[\delta(1+\hat{\lambda}_n) + \Pi^n_{j-1}(K^c_{\delta j-1})] \\
&\leq \delta(1+\hat{\lambda}_n)\sum_{k=2}^{j}\prod_{i=k}^{j}(c_1 e^{c_2|\varepsilon_i|}) + \prod_{i=2}^{j}(c_1 e^{c_2|\varepsilon_i|})\Pi^n_1(K^c_\delta) \\
&\leq \delta(1+\hat{\lambda}_n)\sum_{k=2}^{j}\prod_{i=k}^{j}(c_2 e^{c_1|\varepsilon_i|}) + \delta(1+\hat{\lambda}_n)\prod_{i=1}^{j}(c_1 e^{c_2|\varepsilon_i|})\int \nu[0,1]\pi_0(d\nu),
\end{aligned}
$$

since $h(\cdot)$ is bounded, where c_1 and c_2 are finite constants. So,

$$
\mathbb{E}[\Pi^n_j(K^c_\delta)] \leq c_j\delta.
$$

Let $\delta \to 0$, we have that, for almost all ω,

$$
\Pi^n_j\{\mu;\, \mu([0,1]) = \infty\} = 0.
$$

Therefore, the proof of (i) is completed.

(ii) Let H be the Borel set in the product $(u_\cdot, \pi_\cdot)$ path space. For arbitrary integer $c > 0$, define the left shift H_c by $H_c = \{(u_\cdot, \pi_\cdot);\, (u_{c+\cdot}, \pi_{c+\cdot}) \in H\}$. Then

$$
Q^{n,k}_1(H_c) = \frac{1}{k}\sum_{j=1}^{k} I_{H_c}\big(\Psi^n_1(j,\cdot)\big) = \frac{1}{k}\sum_{j=1}^{k} I_H\big(\Psi^n_1(j+c,\cdot)\big),
$$

and hence

$$
\begin{aligned}
\big|Q^{n,k}_1(H_c) - Q^{n,k}_1(H)\big| &= \Big|\frac{1}{k}\sum_{j=c+1}^{k+c} I_H\big(\Psi^n_1(j,\cdot)\big) - \frac{1}{k}\sum_{j=1}^{k} I_H\big(\Psi^n_1(j,\cdot)\big)\Big| \\
&= \Big|\frac{1}{k}\Big(\sum_{j=k+1}^{k+c} - \sum_{j=1}^{c}\Big) I_H\big(\Psi^n_1(j,\cdot)\big)\Big| \leq \frac{2c}{k} \to 0,
\end{aligned}
$$

as $k \to \infty$ for all ω, c, n, H. In addition to the weak convergence, it yields the stationarity of their limited processes $\big(U^\omega_\cdot, \Pi^\omega_\cdot\big)$ for almost all ω.

(iii) Let $\phi_r(\cdot), \varphi_p(\cdot)$ be two arbitrary finite collections of bounded and continuous real-valued functions. Let c_i be a finite collection of arbitrary integers. For arbitrary integers m, k and bounded and continuous function $q(\cdot)$, define the function

$$
\begin{aligned}
f(\psi_\cdot) =& q\big(\langle u_{c_i}, \phi_r\rangle, \langle \pi_{c_i}, \varphi_p\rangle, b_{c_i};\, c_i \leq m, r, p\big) \\
&\times \Big[\exp\Big\{\sum_{l=m}^{m+k}\langle u_l, \phi_l\rangle(b_{l+1} - b_l)\Big\} - \exp\Big\{\frac{1}{2}\sum_{l=m}^{m+k}\langle u_l, \phi_l\rangle^2\Big\}\Big].
\end{aligned}
$$

We need to show that, for almost all ω,

$$
\int f(\psi_\cdot) Q^\omega(d\psi) = 0. \tag{16}
$$

This, in turn, implies that (for almost all ω) the ε_l^ω are mutually independent $\mathcal{N}(0,1)$ random variables, and are independent of the $U_\cdot^\omega$ and of the "past" of the $\Pi_\cdot^\omega$ process. To get (16), it is enough to show that as $n, k \to \infty$,

$$\mathbb{E}\Big[\int f(\psi_\cdot) Q^{n,k}(d\psi)\Big]^2 = \mathbb{E}\Big[\frac{1}{k}\sum_{j=1}^{k} f\big(\Psi^n(j, c_i)\big)\Big]^2 \to 0.$$

In fact, by the definition of $\Psi^n(j,\cdot)$,

$$\begin{aligned} f\big(\Psi^n(j,c_i)\big) &= q\big(\langle U_{j+c_i}, \phi_j\rangle, \langle \Pi^n_{j+c_i}, \varphi_p\rangle, B_{j+c_i} - B_j; c_i \le m, j, p\big) \\ &\times \Big[\exp\Big\{\sum_{l=m}^{m+k} \langle U_{j+l}, \phi_l\rangle (B_{j+l+1} - B_{j+l})\Big\} - \exp\Big\{\frac{1}{2}\sum_{l=m}^{m+k} \langle U_{j+l}, \phi_l\rangle^2\Big\}\Big]. \end{aligned}$$

Using the fact that $B_{j+l+1} - B_{j+l} = \varepsilon_{j+l} \sim \mathcal{N}(0,1)$ are mutually independent among all the j, l, so one can see that the mean square converges to 0.

(iv) Write $|z|_1^2 = \min\{|z|^2, 1\}$. To identify $Y_\cdot^\omega$, it is necessary to prove for each integer s,

$$\mathbb{E}\int Q^\omega(d\psi)\big|y_s - h\big(\langle u_s, \phi_0\rangle\big) - (b_{s+1} - b_s)\big|_1^2 = 0. \tag{17}$$

which implies (iv) for almost all ω. This can be shown by using the definition of occupation measures and the weak convergence. Note that

$$\begin{aligned} &\mathbb{E}\int Q^{n,k}(d\psi)\big|y_s - h\big(\langle u_s, \phi_0\rangle\big) - (b_{s+1} - b_s)\big|_1^2 \\ &= \mathbb{E}\frac{1}{k}\sum_{j=1}^{k}\Big|Y_{j+s} - h\big(\langle U_{j+s}, \phi_0\rangle\big) - (B_{j+s+1} - B_{j+s})\Big|_1^2 = 0. \end{aligned}$$

Then, since the integrand in (17) is a bounded and continuous function of $\psi_\cdot$ and $Q^{n,k}$ weakly converges to Q, equality (17) follows.

(v) Adopt the notations in (iii). Using the similar arguments in Budhiraja and Kushner (1999), we can show that, for any real number x,

$$\int Q^\omega(d\psi) q(\langle u_{c_i}, \phi_r\rangle; c_i \le m, r)\Big[e^{ix\langle u_{m+1}, \phi\rangle} - \int e^{ix\langle \mu, \phi\rangle}\mathbb{P}(d\mu, 1|u_m)\Big] = 0,$$

where $\mathbb{P}(d\mu, 1|x)$ is the one step transition function of $U_\cdot$. This implies that $U_\cdot^\omega$ is Markov with the transition function of $U_\cdot$.

(vi) For arbitrary canonical sample path $\psi_\cdot$ and integer m, define a function

$$A(\psi_m) = \langle \pi_m, \phi\rangle - \frac{\mathbb{E}_{\{\pi_{m-1}, y_m\}}\{\phi(\tilde{U}_1) R(\tilde{U}_1, y_m)\}}{\mathbb{E}_{\{\pi_{m-1}, y_m\}}\{R(\tilde{U}_1, y_m)\}}.$$

We are to prove that, for almost all ω and all m, $A(\Psi_m^\omega) = 0$, with probability 1 which implies (15). This can be done by showing that

$$0 = \mathbb{E}\int Q^\omega(d\psi)\big|A(\psi_m)\big|_1^2. \tag{18}$$

By the weak convergence, the prelimit form of the right side of (18) is

$$\mathbb{E}\int Q^{n,k}(d\psi)\big|A(\psi_m)\big|_1^2 = \frac{1}{k}\sum_{j=1}^{k}\mathbb{E}\big|A(\Psi^n(j,m))\big|_1^2.$$

Thus, to obtain (18) it suffices to show that $\mathbb{E}\big|A(\Psi^n(j,\cdot)\big|_1^2 \to 0$ uniformly in j as $n \to \infty$. Actually, by Definition (11), Theorem 5 and the continuity of ϕ, ϕ_0, h, we have

$$\begin{aligned}
\mathbb{E}\big|A(\Psi^n(j,\cdot))\big|_1^2 &= \mathbb{E}\Big|\langle \Pi_j^n, \phi\rangle - \frac{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{\phi(\tilde{U}_1)R(\tilde{U}_1, Y_j)\}}{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{R(\tilde{U}_1, Y_j)\}}\Big|_1^2 \\
&= \mathbb{E}\Big|\frac{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{\phi(\tilde{U}_1^n)R(\tilde{U}_1^n, Y_j)\}}{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{R(\tilde{U}_1^n, Y_j)\}} \\
&\quad - \frac{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{\phi(\tilde{U}_1)R(\tilde{U}_1, Y_j)\}}{\mathbb{E}_{\{\Pi_{j-1}^n, Y_j\}}\{R(\tilde{U}_1, Y_j)\}}\Big|_1^2 \to 0,
\end{aligned}$$

uniformly in j as $n \to \infty$. Hence, (18) holds for any weak sense limit.

□

Remark 3 Theorem 7 yields that $\{Q_1^{n,k}(\cdot)\}$ converges weakly to the probability measure of $\{U_\cdot, \Pi_\cdot\}$ along the chosen subsequence. Keeping in mind that $\rho(\cdot)$ is the unique invariant measure of $\{U_\cdot, \Pi_\cdot\}$, so one can assert $\{Q_1^{n,k}(\cdot)\}$ converges weakly to $\rho(\cdot)$ along any (n,k). Formula (12) is therefore derived. Particularly, for either $\phi \in C_b(\mathcal{X})$ or Dirac delta function, we have

$$\lim_{n,k\to\infty} \frac{1}{k}\sum_{j=1}^k \mathbb{E}\langle \Pi_j^n, \phi\rangle^2 = \int_{\mathcal{X}\times\mathcal{P}(\mathcal{X})} \mathbb{E}\langle \alpha, \phi\rangle^2 \rho(d\nu, d\alpha) = \lim_{k\to\infty}\frac{1}{k}\sum_{j=1}^k \mathbb{E}\langle \Pi_j, \phi\rangle^2.$$

Finally we are ready to conclude our main result.

Theorem 8 *For all bounded and continuous real-valued function ϕ,*

$$\frac{1}{k}\sum_{j=1}^k \big|\langle \Pi_j^n, \phi\rangle - \langle \Pi_j, \phi\rangle\big|^2 \to 0 \tag{19}$$

in probability as $n, k \to \infty$.

Proof Thanks to Remark 3, we have

$$\begin{aligned}
&\lim_{n,k\to\infty} \mathbb{E}\frac{1}{k}\sum_{j=1}^k \big|\langle \Pi_j^n, \phi\rangle - \langle \Pi_j, \phi\rangle\big|^2 \\
&= \lim_{n,k\to\infty}\Big(\frac{1}{k}\sum_{j=1}^k \mathbb{E}\langle \Pi_j^n, \phi\rangle^2 + \frac{1}{k}\sum_{j=1}^k \mathbb{E}\langle \Pi_j, \phi\rangle^2 - \frac{2}{k}\sum_{j=1}^k \mathbb{E}\langle \Pi_j^n, \phi\rangle\langle \Pi_j, \phi\rangle\Big) \\
&\overset{*}{=} \lim_{k\to\infty}\frac{2}{k}\sum_{j=1}^k \mathbb{E}\langle \Pi_j, \phi\rangle^2 - \lim_{n,k\to\infty}\frac{2}{k}\sum_{j=1}^k \mathbb{E}\Big(\langle \Pi_j^n, \phi\rangle \mathbb{E}\big(\phi(U_j)|\mathcal{F}_j^Y\big)\Big) \\
&= \lim_{k\to\infty}\frac{2}{k}\sum_{j=1}^k \mathbb{E}\Big(\langle \Pi_j, \phi\rangle \mathbb{E}\big(\phi(U_j)|\mathcal{F}_j^Y\big)\Big) - \lim_{n,k\to\infty}\frac{2}{k}\sum_{j=1}^k \mathbb{E}\Big(\langle \Pi_j^n, \phi\rangle \mathbb{E}\big(\phi(U_j)|\mathcal{F}_j^Y\big)\Big) \\
&= 0,
\end{aligned}$$

where the equality (*) holds by formula (12). This implies the convergence in probability. □

4 Simulation results

To specifically detect the source of pollution and its concentration along the river, here we set dispersion $a = 1$, velocity $b = 2$, dilution rate $\alpha = 5$, $h(z) = 3z$, observation location $x_0 = 0.2$, the initial condition $U_0(y) = y$. The state process $U_\cdot$ is supposed to be tested against ϕ_0. Let ϕ_0 be a Dirac delta function with $\phi_0(y) = \delta(y - x_0)$, indicating that the observation data are exactly the chemical concentration at station $x_0 = 0.2$. Then the observation become

$$\begin{aligned} Y_i =& h(\langle U_i, \phi_0\rangle) + \varepsilon_i \\ =& \int_0^1 3\delta(y - x_0) U_i(y) dy + \varepsilon_i \\ =& 3U_i(x_0) + \varepsilon_i, \end{aligned}$$

with $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Write $D(t, y) = e^{-\alpha t} P_t(y, x_0)$, where the Green's function

$$\begin{aligned} P_t(y, x) = \frac{2}{e^2 - 1} + \sum_{j=1}^{\infty} 2e^{-(1+j^2\pi^2)t - x - y} \sin(j\pi x + \tan^{-1}(j\pi)) \\ \times \sin(j\pi y + \tan^{-1}(j\pi)). \end{aligned}$$

To simulate signal process U_t, suppose the undesired chemical are released from location $\beta_0 = 0.6$ by a Poisson process with intensity $\lambda_0 = 10$. One can generate a series of arrival time $0 < \tau_1 < \tau_2 < \cdots \leq T = 20$ of the Poisson point process $N_{\lambda_0 t}$ by matlab. Then we are able to calculate the true signal at observation station:

$$\begin{aligned} U_i(x_0) =& e^{-\alpha t_i} \int_0^1 P_{t_i}(y, x_0) U_0(dy) + \sum_{\tau_k \leq t_i} e^{-\alpha(t_i - \tau_k)} P_{t_i - \tau_k}(\beta_0, x_0) \\ \triangleq& A(t_i) + \sum_{\tau_k \leq t_i} D(t_i - \tau_k, \beta_0). \end{aligned}$$

According to Theorem 2, define

$$m_k(n) = \frac{1}{n} \sum_{i=1}^{n} Y_i^k = \frac{1}{n} \sum_{i=1}^{n} \Big(3A(t_i) + 3 \sum_{\tau_k \leq t_i} D(t_i - \tau_k, \beta_0) + \varepsilon_i\Big)^k, \ k = 1, 2,$$

and

$$\begin{aligned} g_1(\theta) =& \int_{\mathcal{X}} h(\langle \nu, \phi_0\rangle) F(d\nu) = 3\lambda \int_0^\infty D(s, \beta) ds, \\ g_2(\theta) =& \int_{\mathcal{X}} h^2(\langle \nu, \phi_0\rangle) F(d\nu) \\ =& 9\lambda \int_0^\infty D^2(s, \beta) ds + 9\lambda^2 \left(\int_0^\infty D(s, \beta) ds\right)^2. \end{aligned}$$

Assume that the detection time length $T = 20$, sampling time frequency $\Delta = 0.01$. In the calculation of $P_t(y, x)$, we will take the addition up to $j = 100$. The continuous integral $A(t_i)$ will be approximated by numerical integration with partition $\Delta_y = 0.001$.

Remark 4 In this example, we can verify the existence of the partial derivatives required in Theorem 2. However, it is hard to prove $det(J) \neq 0$ because of the complexity of the

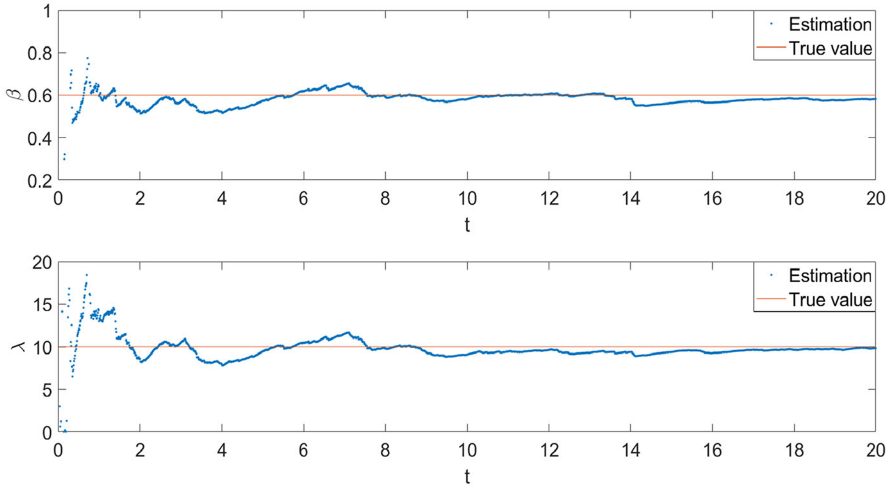

Fig. 1 Estimation of parameters

summations in the expressions of $P_s(\beta, x_0)$. For this case, we have checked on

$$det(J(\theta)) = \begin{vmatrix} \partial g_1(\theta)/\partial\beta & \partial g_1(\theta)/\partial\lambda \\ \partial g_2(\theta)/\partial\beta & \partial g_2(\theta)/\partial\lambda \end{vmatrix} \neq 0$$

in $\{(\beta, \lambda); 0 < \beta < 1, 1 < \lambda \leq 500, \Delta_\beta = 0.001, \Delta_\lambda = 0.01\}$ numerically.

Let

$$f_1^n(\theta) = g_1(\theta) - m_1(n) = 0, \quad f_2^n(\theta) = g_2(\theta) + 1 - m_2(n) = 0.$$

Given the initial guess $\hat{\beta}_0 = 1/3$, $\hat{\lambda}_0 = 5$, for the n-th ($n = 1, \cdots, 2000$) sampling, with Newton–Raphson formula

$$[\beta_{k+1}, \lambda_{k+1}] = [\beta_k, \lambda_k] - [f_1^n(\beta_k, \lambda_k), f_2^n(\beta_k, \lambda_k)] \begin{bmatrix} \partial f_1^n/\partial\beta_k & \partial f_2^n/\partial\beta_k \\ \partial f_1^n/\partial\lambda_k & \partial f_2^n/\partial\lambda_k \end{bmatrix}^{-1},$$

we can get an approximation for the root of $f_1^n = f_2^n = 0$, denoted by $(\hat{\beta}_n, \hat{\lambda}_n)$. Simulation results for $(\hat{\beta}_n, \hat{\lambda}_n)$ are shown in Fig. 1.

Now we are to evaluate the concentration $U.$ by constructing its approximated filter. Taking $n = 550$, the estimators are $\hat{\beta}_{550} = 0.597$, $\hat{\lambda}_{550} = 10.038$. Suppose $\phi(z) = z$, then $\langle \Pi_j, \phi \rangle = \mathbb{E}\{U_j \mid \mathcal{F}_j^Y\}$. Let $M = 2000$, with formula (11) the approximated filter Π_j^n by the sample average is defined as

$$\langle \Pi_j^n, \phi \rangle = \frac{\sum_{l=1}^M \tilde{U}_j^{n,l} R(\tilde{U}_j^{n,l}, Y_j)}{\sum_{l=1}^M R(\tilde{U}_j^{n,l}, Y_j)},$$

where

$$\begin{aligned} \tilde{U}_j^{n,l} &= e^{-\alpha t_1} \int_0^1 P_{t_1}(y, x_0) \tilde{U}_{j-1}^{n,l}(dy) + \int_{t_{j-1}}^{t_j} e^{-\alpha(t_j - s)} P_{t_j - s}(\hat{\beta}_{550}, x_0) d\tilde{N}^l_{\hat{\lambda}_{550} s} \\ &= e^{-\alpha t_1} \int_0^1 P_{t_1}(y, x_0) \tilde{U}_{j-1}^{n,l}(dy) + \sum_{t_{j-1} < \hat{\tau}_k^l \leq t_j} e^{-\alpha(t_j - \hat{\tau}_k^l)} P_{t_j - \hat{\tau}_k^l}(\hat{\beta}_{550}, x_0), \end{aligned}$$

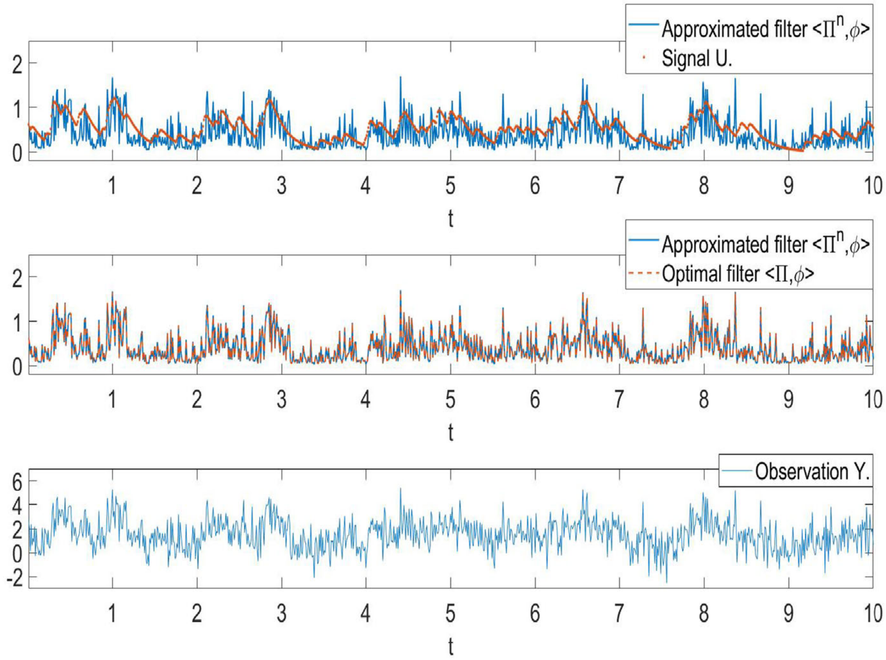

Fig. 2 Approximation of signal process

$$
\begin{aligned}
R(\tilde{U}_j^{n,l}, Y_j) &= \exp\left\{Y_j h(\langle \tilde{U}_j^{n,l}, \phi_0\rangle) - \frac{1}{2}h^2(\langle \tilde{U}_j^{n,l}, \phi_0\rangle)\right\} \\
&= \exp\left\{3Y_j \cdot \tilde{U}_j^{n,l}(x_0) - \frac{9}{2}(\tilde{U}_j^{n,l}(x_0))^2\right\},
\end{aligned}
$$

and $\{\hat{\tau}_k^l, l = 1, \cdots, M\}$ is a series of jumping time generated by Poisson process with intensity $\hat{\lambda}_{550}$, the initial $U_0^{n,l}(dy) = U_0(dy)$. Replacing $\hat{\beta}_{550}, \hat{\lambda}_{550}$ by the true parameters β_0, λ_0, one can obtain the optimal filter $\langle \Pi, \phi\rangle$ with the same definition. Finally, the approximation $\langle \Pi^n, \phi\rangle$, optimal filter $\langle \Pi, \phi\rangle$, true concentration $U_.$ and observation data $Y_.$ at location x_0 are illustrated in Fig. 2. The time length in Fig. 2 is reduced to $T = 10$ just for the sake of clarity.

The upper plot in Fig. 2 shows that the computed approximating filter points always fall near the true signal. It can be seen from the middle plot that the approximated filter almost overlaps the optimal filter. By calculation, the order of magnitude of pathwise average error $r = \frac{1}{k}\sum_{j=1}^k \left|\langle \Pi_j^n, \phi\rangle - \langle \Pi_j, \phi\rangle\right|^2$ which is introduced by Theorem 8 is equal to -4 actually. And the error is getting closer to 0 as time goes by, for $r = 1.65 \times 10^{-4}$ when $k = 100$; $r = 1.47 \times 10^{-4}$ when $k = 200$; $\cdots$; $r = 1.23 \times 10^{-4}$ when $k = 900$; $r = 1.16 \times 10^{-4}$ when $k = 1000$.

5 Concluding remarks

Under the background of environmental pollution detection, we put forward a nonlinear filtering problem over a long time interval where the signal process was driven by an SPDE involved with Poisson Process and unknown parameters. According to the discrete samples, the unknown parameters were figured out by the ergodic theory and statistical schemes at first. Then we proceeded to verify the uniqueness of invariant measure for the signal-filter process. Since the optimal filter is considerably hard to construct, the computed approximating filter was established. The convergence properties were finally argued both by theory and matlab simulations. This paper provides a new way to solve a one-dimensional nonlinear filtering problem with unknown parameters. The statistical method can be adopted to the cases with more unknown parameters, which is likely to happen in practice. It could also be extended to the cases that pollutants are issued at multiple waste outfalls with various random magnitudes. Besides that, the approach in this model can be developed to study other problems such as the multi-dimensional case of atmospheric pollution problem.

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Data availability Enquiries about data availability should be directed to the authors.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Human and animal rights This article does not contain any studies with human participants or animals performed by any of the authors.

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